

Md Showkat Hossain Chy

Curriculum vitae

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Seven years on distributed systems and machine learning over large graphs, with earlier work on firmware running under fixed memory and power budgets. The common thread is correctness under concurrency, failure, and scale.

Seattle, Washington, USA — U.S. Permanent Resident — mdshowkathossainchy@gmail.com

Research record

12	148	8
peer-reviewed papers	citations	h-index

Venues include IEEE/ACM ASE, ECSA, CLOSER, Journal of Systems and Software, Cluster Computing, Electronics, Telecom, SeGAH, and ICSEC.

Education

Jan 2022 - Dec 2023	Baylor University M.S. in computer science, Waco, Texas Concentration in machine learning for software systems, microservice architecture, and distributed data processing.
Jan 2013 - May 2018	Daffodil International University B.Sc. in computer science and engineering, Dhaka, Bangladesh

Experience

May 2026 - present	Founder and Engineer Stealth AI startup, Seattle, Washington Building an agentic AI platform that monitors municipal records, council agendas, zoning dockets, and permit filings across U.S. cities, and converts them into early business signals for commercial real estate and multi-site operators.
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Jan 2025 - Apr 2026

Software Development Engineer, AWS Infrastructure Supply Chain

Amazon Web Services, Seattle, Washington

Product lifecycle management system governing hardware parts and engineering changes across AWS data centres and device manufacturing.

Modelled millions of hardware components, assemblies, and bills of materials as a graph in Amazon Neptune and DynamoDB, supporting change-impact traversal across products, sites, and programs.

Held the hardware system of record to a 24-hour data-freshness SLA with an event-driven reconciliation service on EventBridge, SQS, AWS Batch, and DynamoDB that detected and safely replayed dropped, duplicated, and out-of-order updates.

Built the platform end to end: a Java and Spring Boot rules engine evaluating site, region, and program policy from declarative configuration, through a GraphQL, React, and TypeScript interface. 60% gain in hardware qualification efficiency, with new categories onboarded across 100+ manufacturing sites without per-class code.

Integrated 15+ cross-region systems against a single source of product data behind versioned Spring Boot REST APIs and events, with role-based access control, fine-grained access control, and per-caller rate limits.

Raised integration test coverage from 60% to 90% with Cypress, and instrumented the platform with CloudWatch, real user monitoring, CodeGuru profiling, and Lambda-based TLS certificate monitoring.

Made attributes, classification, and lifecycle rules data-driven, so new hardware categories onboarded across 100+ manufacturing sites with no engineering work and no custom code per product class.

Established team-wide AI-assisted development workflows using Claude Code and Amazon Q.

Mar - Dec 2024

Research Associate II, Software Engineering

CloudHubs Lab, University of Arizona, Tucson, Arizona

Led the lab's graph-learning line: graph neural networks and variational autoencoders with fuzzy c-means clustering over large static-analysis dependency graphs, recovering latent structure automatically. Published at IEEE/ACM ASE 2024 and in Cluster Computing.

Designed and released a multi-variant benchmark suite giving controlled, reproducible subjects for architecture research.

Characterised how changes propagate across service boundaries.

Mentored 15 graduate and undergraduate students in experimental design and empirical evaluation.

Mar - Dec 2024

Full-Stack Developer, International Collegiate Programming Contest

ICPC Foundation, University of Arizona, Tucson, Arizona

Contest platform for the world's largest collegiate programming competition.

Built contest-management workflows in Spring Boot, React, PostgreSQL, and REST covering registration, scheduling, and logistics for 50,000+ students annually across 50+ regions.

Designed a multi-tenant, configuration-driven platform letting regional managers publish events independently. 45% reduction in administrative time, \$12K saved annually.

Ran static analysis with SonarQube, resolving 1,600+ defects and security issues.

Covered team registration, contest scheduling, hotel logistics, and event administration, replacing the one-off websites each region had been maintaining.

May - Aug 2023

Software Engineer, Summer Intern

Tyson Foods, Springdale, Arkansas

Delivered Unreal Engine virtual-reality quality-inspection training for 200+ manufacturing personnel, worth \$233K in annual savings.

Automated 3D asset generation, cutting a two-week manual process to 10 minutes on an AWS EC2 and Perforce pipeline.

Jan 2022 - Dec 2023

Graduate Research and Teaching Assistant, Computer Science

Baylor University, Waco, Texas

Built distributed MapReduce and PySpark pipelines on Amazon EMR over 10M+ blockchain transactions, extracting opcode and control-flow features from Solidity bytecode and training anomaly-detection models. 65% improvement in analysis efficiency.

Ran a controlled empirical evaluation of four JVM message queues (Kafka, Artemis, Pulsar, RocketMQ), measuring latency, message ordering, and delivery reliability under concurrency.

Implemented a transaction manager in Python enforcing ACID guarantees through strict two-phase locking and write-ahead logging, with crash recovery under concurrent workloads.

Taught software engineering (Java) and C++ programming: 125 students across five sections.

Jul 2018 - Dec 2021

Software Engineer, Wearable Platform

Samsung Research, Dhaka, Bangladesh

Wearable connectivity and firmware for Galaxy Watches, fitness bands, Samsung Health, and the Smart Ring program.

Developed FreeRTOS firmware and device drivers in C and C++ over SPI, UART, and I2C for heart-rate, stress, and sleep tracking across 90+ exercise modes. Profiled and debugged races and timing defects at the microcontroller level with JTAG and Trace32 under fixed memory and power budgets.

Designed the Bluetooth Low Energy and GATT control and data planes — service discovery, characteristics, message framing, connection stability — sustaining real-time sensor streaming on a 15 ms cadence.

Founding engineer on phone-to-wearable connectivity shipped to 2M+ devices with no known production defects. Authored the Samsung Accessory Protocol libraries in C and C++ for Android and iOS.

Built rollback-safe over-the-air update workflows with recovery paths for devices that could not be physically retrieved: 95% test coverage, 99% rollout success, zero bricked devices.

Defined the latency and throughput targets the connectivity stack was held to, and worked against a Linux toolchain with emWin on the display side.

Jan - Jun 2018

Software Engineer

Divine IT Limited, Dhaka, Bangladesh

Built Django, DRF, and PostgreSQL ERP modules across payroll, HR, inventory, sales, and procurement for 5,000+ daily users, improving throughput by 30%.

Technical range

Languages

C, C++, Java, Python, TypeScript, JavaScript, SQL, Bash

Distributed and cloud

Event-driven architecture, microservices, data modelling, MapReduce, PySpark, Amazon EMR, Kafka, Pulsar, Artemis, RocketMQ, DynamoDB, Amazon Neptune, PostgreSQL, MySQL, Redis, Docker, Kubernetes, and AWS (Lambda, EventBridge, SQS, SNS, Batch, EC2, S3, IAM, API Gateway, CloudFormation, CDK)

Machine learning	Graph neural networks, variational autoencoders, representation learning over graphs, clustering, anomaly detection, feature engineering, PyTorch, TensorFlow
Embedded and firmware	FreeRTOS, microcontroller firmware, device drivers over SPI, UART, and I2C, Bluetooth Low Energy and GATT, rollback-safe over-the-air updates, JTAG and Trace32 profiling, emWin, memory debugging, low-power systems, wireless protocol design
Engineering practice	JUnit, Cypress, integration and regression testing, coverage analysis, test-driven development, CI/CD, SonarQube, static analysis, Spring Boot, GraphQL, REST, observability with CloudWatch, Grafana, and CodeGuru, Git, Perforce

Publications

Journal articles

Multivocal Study on Microservice Dependencies

A. S. Abdelfattah, T. Cerny, **M. S. H. Chy**, M. A. Uddin, S. Perry, C. Brown, J. Yero, K. Sooksatra

JSS 2025 volume 222, article 112334

doi 10.1016/j.jss.2025.112334

Benchmarking Micro2Micro Transformation: An Approach with Graph Neural Networks and Variational Autoencoders

M. S. H. Chy, K. Sooksatra, J. Yero, T. Cerny

Cluster Computing 2024 volume 27, number 4, pages 4171–4185

doi 10.1007/s10586-024-04526-z

Comparative Evaluation of Java Virtual Machine-Based Message Queue Services: A Study on Kafka, Artemis, Pulsar, and RocketMQ

M. S. H. Chy, M. A. R. Arju, S. M. Tella, T. Cerny

Electronics 2023 volume 12, article 4792

doi 10.3390/electronics12234792

Benchmarking Message Queues

R. Maharjan, **M. S. H. Chy**, M. A. R. Arju, T. Cerny

Telecom 2023 volume 4, number 2, pages 298–312

doi 10.3390/telecom4020018

Conference papers

Using Static Analysis to Aid Monolith to Microservice System Transformation: Tuning Fuzzy c-Means in a VAE-Based GNN Approach

K. Sooksatra, **M. S. H. Chy**, M. A. R. Arju, T. Cerny, P. Rivas

IEEE/ACM ASE 2024

doi 10.1145/3691621.3694933

A Multi-variant Benchmark for Microservice Systems in Software Engineering Research

T. Cerny, **M. S. H. Chy**, M. A. R. Arju, K. Sooksatra, A. S. Abdelfattah, V. Lenarduzzi

ECSA 2024 pages 21–29

doi 10.1007/978-3-031-71246-3_3

On Maintainability and Microservice Dependencies: How Do Changes Propagate?

T. Cerny, **M. S. H. Chy**, A. S. Abdelfattah, J. Soldani, J. Bogner

CLOSER 2024 volume 1, pages 277–286

doi 10.5220/0012725200003711

On Dependencies in Microservices: Dependency Management and Maintainability

T. Cerny, **M. S. H. Chy**, M. A. Uddin, A. S. Abdelfattah, J. Soldani, J. Bogner

Springer CCIS

2024 pages 29–52

doi 10.1007/978-3-032-17286-0_2

Evolution and Anti-patterns Visualized: MicroProspect in Microservice Architecture

L. Adams, A. Elsayed, **M. S. H. Chy**, S. Perry, P. Harris, T. Cerny, D. Amoroso d'Aragona, D. Taibi

ECSA

2023 pages 309–325

doi 10.1007/978-3-031-66326-0_19

A Systematic Review on Software Design Patterns in Today's Perspective

M. Rahman, **M. S. H. Chy**, S. Saha

IEEE SeGAH

2023 Athens, Greece, pages 1–8

Bless: An Interactive Application to Show Gratitude towards Doctors

M. Rahman, M. S. Rahaman, **M. S. H. Chy**

IEEE SeGAH

2023 Athens, Greece, pages 1–7

Exploring Chromatin Interaction Between Two Human Cell Types and Different Normalization Techniques for Hi-C Data

M. Rahman, **M. S. H. Chy**, A. Rahman, S. Ahmed, T. Sharmin, M. S. Rahaman

ICSEC

2023 Thailand, pages 53–59

Academic service and peer review

Program chair, 32nd International Conference on Information Systems Development, 2024.

Program chair, 18th IEEE International Conference on Service-Oriented System Engineering, 2024.

Reviewer, Springer Cluster Computing, 2024.

Member, IEEE, IEEE Computer Society, and IEEE Young Professionals, 2024 to present.

Teaching and mentoring

Teaching assistant, software engineering (Java) and C++ programming, Baylor University — 125 students across five sections, 2022–2023.

Research mentor, CloudHubs Lab, University of Arizona — 15 graduate and undergraduate students in software-architecture research and empirical methods, 2024.

Capstone supervisor, Baylor University, from requirements through final delivery.

Engineering mentor, Samsung Research — four junior engineers in debugging, test-driven development, and code review, 2018–2021.

Contest trainer, judge, and problem setter, Daffodil International University, 2015–2017.

Competitive programming and awards

ACM ICPC contestant, 2013–2016 — four Dhaka Site Regionals, six national contests, around 1,200 problems solved.

First runner-up, Inter-University Programming Contest, 2014. Champion, DIIT Code Fest, 2015.

Chief Technology Officer Award, Samsung Electronics, for the Galaxy Fit project, 2019.

Languages

English — IELTS overall band 7.5, listening 8.5, reading 8.0.

Bengali — native proficiency.